DG Interns Hub — Week 3 Final Report

Linux Logs, Sysmon, auditd and Snort

Student: Ghanisth Mittal
Environment: Kali Linux + Metasploitable 2 on VMware
Focus: SOC monitoring, endpoint telemetry, Linux auditing, IDS detection and log analysis

1. Week 3 Overview

Week 3 focused on collecting and analyzing security telemetry, configuring endpoint and network monitoring tools, creating Snort detection rules, detecting reconnaissance activity, and practicing SOC-style alert investigation.

Lab environment rationale

The assigned environment referenced Kali, Ubuntu and Windows. For the network-security practical, Kali Linux was used as the analyst/attacker workstation and Metasploitable 2 as the intentionally vulnerable target. This was selected because the available VMware lab already provided an isolated, authorized Kali + Metasploitable setup suitable for service discovery and IDS validation. Windows was used separately for Sysmon endpoint logging.

2. Completed Work

Area	Completed work	Evidence
Theory	10 Q&A; documents covering EDR, Sysmon, auditd, Snort, detection methods, T1046, Linux logs and log ingestion.	Theory Q&A;
Sysmon	Event IDs 1, 3 and 11; Operational Log and user-creation evidence.	Sysmon screenshots
auditd	Service verification; EXECVE command execution; USER_AUTH authentication evidence.	auditd screenshots
Snort	Snort 3 validation; custom ICMP, SSH, HTTP and T1046 rules; alert testing.	Snort screenshots
T1046	Nmap SYN scan detected with custom Snort rule.	T1046 alert screenshot
Training	4 TryHackMe rooms and 6 LetsDefend lessons completed.	Completion screenshots
Cheat sheet	SOC security monitoring quick reference created.	Week_3_SOC_Cheat_Sheet.docx

3. auditd Practical Evidence

The screenshots below are the strongest auditd evidence supplied for the report. They show the service state, command execution telemetry, command ID/event data and USER_AUTH authentication evidence.

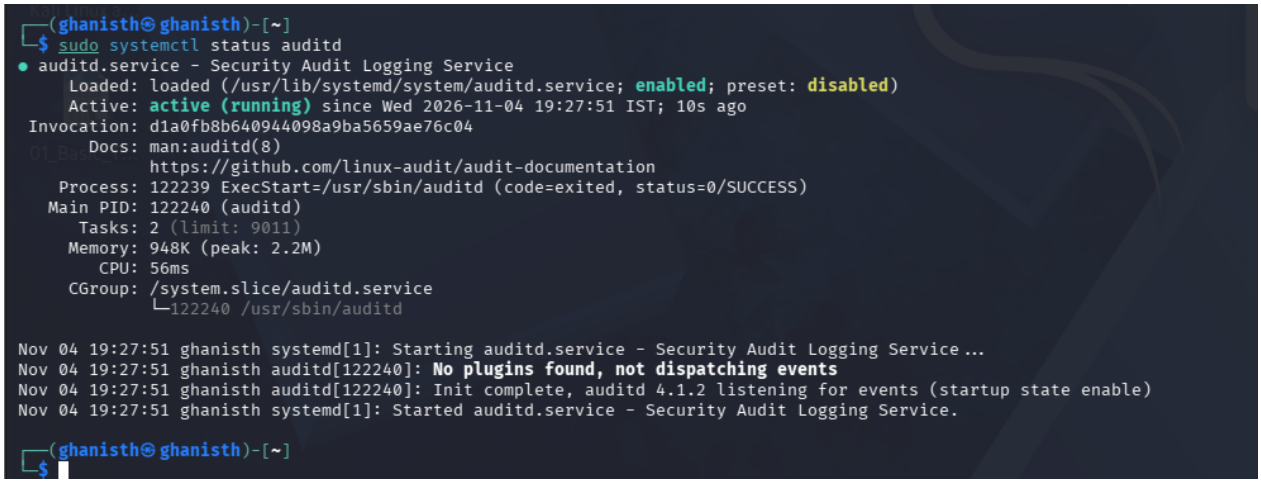


Figure 1. auditd service status — active (running).



Figure 2. auditd EXECVE evidence showing command execution events and the Week 3 audit key.

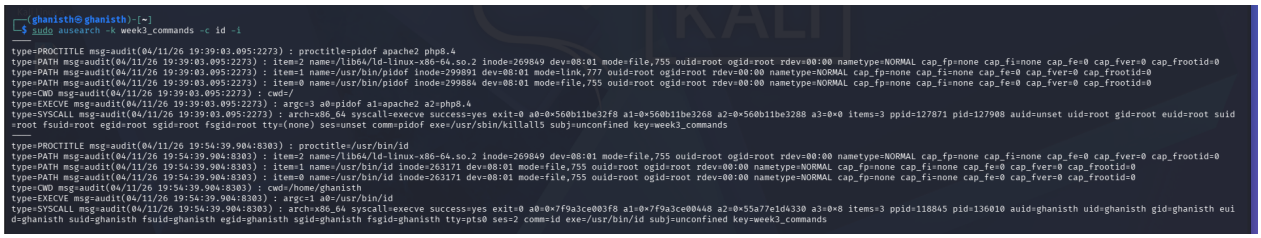


Figure 3. auditd command/event evidence.



Figure 4. auditd USER_AUTH evidence showing successful PAM authentication through /usr/bin/su.

Note: USER_LOGIN returned no matching event during the tested period, so it is not presented as verified evidence.

4. Snort Configuration & Detection Evidence

Snort 3 was configured for the VMware lab subnet. The configuration was tested with the Snort validation command and custom local rules were loaded.

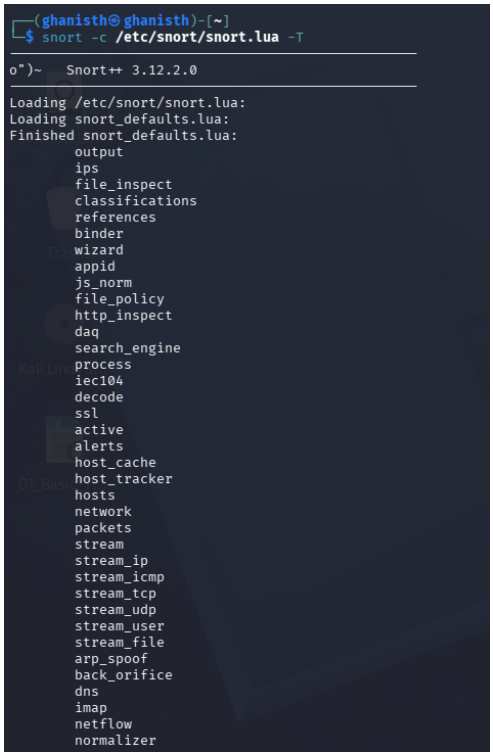


Figure 5. Snort 3 configuration validation — 0 warnings.

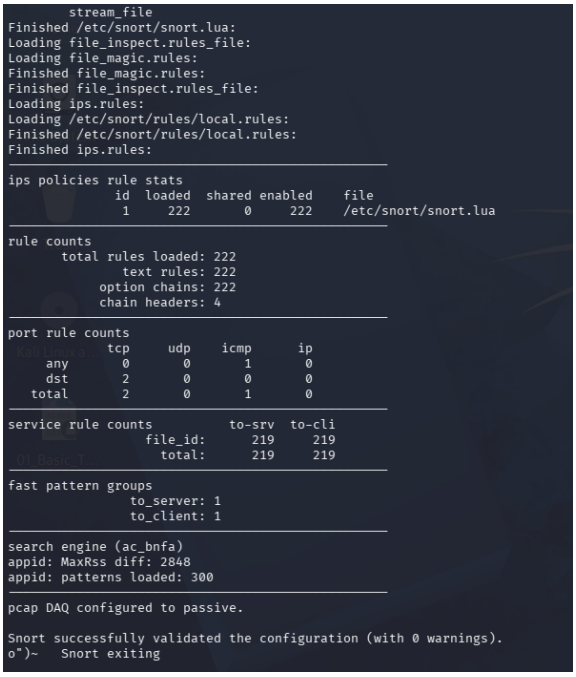


Figure 6. Snort custom rules loaded and validated.

```
┌──(ghanisth㉿ghanisth)-[~]
└─$ sudo snort -A alert_fast -q -c /etc/snort/snort.lua -i eth0
^C11/04-20:31:26.531773 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 192.168.144.128
11/04-20:31:27.529752 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 192.168.144.128
11/04-20:31:28.536231 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 192.168.144.128
11/04-20:31:29.537404 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 8.8.8.8 → 192.168.144.128
```

Figure 7. Verified custom ICMP Ping Detected alert.

```
┌──(ghanisth㉿ghanisth)-[~]
└─$ sudo snort -A alert_fast -q -c /etc/snort/snort.lua -i eth0
[sudo] password for ghanisth:
11/04-20:45:54.517903 [**] [1:1000002:1] "SSH Connection Attempt" [**] [Priority: 0] {TCP} 192.168.144.128:37984 → 192.168.144.129:22
11/04-20:46:41.543242 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.128 → 192.168.144.129
11/04-20:46:41.544024 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.129 → 192.168.144.128
11/04-20:46:42.544692 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.128 → 192.168.144.129
11/04-20:46:42.545341 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.129 → 192.168.144.128
11/04-20:46:43.546136 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.128 → 192.168.144.129
11/04-20:46:43.546746 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.129 → 192.168.144.128
11/04-20:46:44.562145 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.128 → 192.168.144.129
11/04-20:46:44.562818 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.144.129 → 192.168.144.128
11/04-20:46:50.443881 [**] [1:1000002:1] "SSH Connection Attempt" [**] [Priority: 0] {TCP} 192.168.144.128:37168 → 192.168.144.129:22
```

Figure 8. Verified SSH Connection Attempt and ICMP alerts involving Metasploitable 2.

5. Nmap + MITRE ATT&CK; T1046 Evidence

An authorized SYN scan was run against Metasploitable 2 using **sudo nmap -sS -p 1-1000 192.168.144.129**. The custom Snort rule generated repeated alerts with SID 1000004, mapping the activity to MITRE ATT&CK; T1046 — Network Service Discovery.

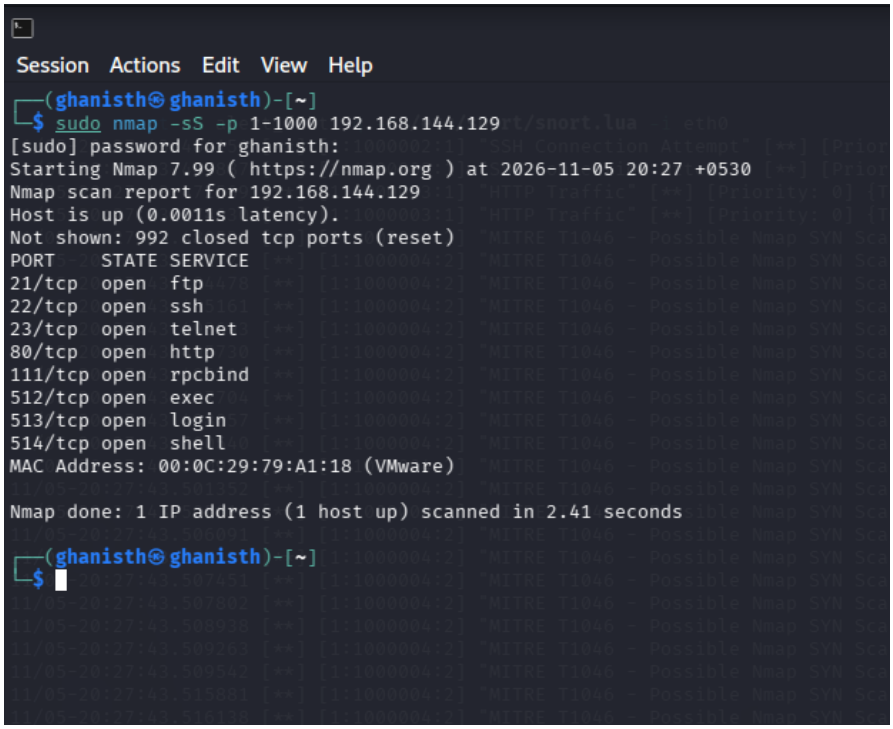


Figure 9. Nmap SYN scan against the authorized Metasploitable 2 target.

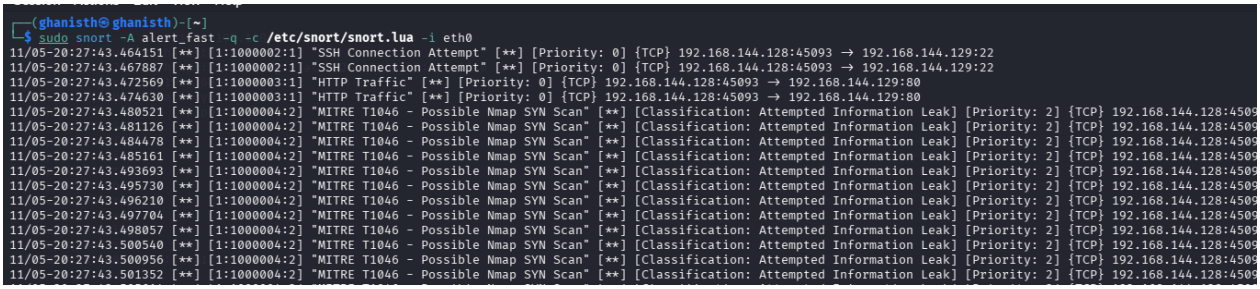


Figure 10. Snort T1046 reconnaissance alerts generated by the Nmap SYN scan.

6. Snort Installation / Network Evidence

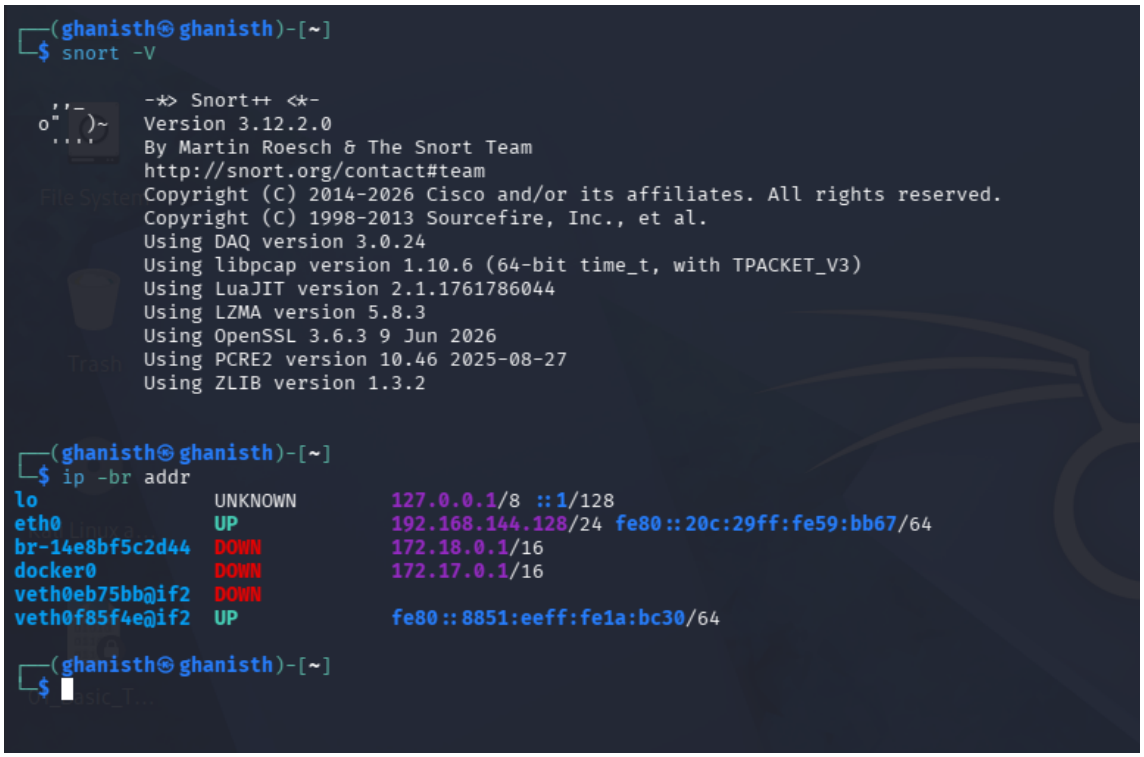


Figure 11. Snort version and Kali network interface evidence.

The screenshot confirms Snort 3.12.2.0 and the Kali eth0 address 192.168.144.128/24 used in the lab.

7. Specific Challenges, Errors and Fixes

Challenge / error	Fix / lesson
Snort 3 rejected the Snort 2-style 'threshold' keyword with an unknown rule keyword error.	Replaced threshold with Snort 3 detection_filter syntax and re-ran validation successfully.
The original ICMP direction did not observe the required return traffic.	Changed the ICMP rule direction from -> to <> and verified an ICMP alert.
Self-SSH on Kali was not useful for remote network-path evidence because loopback traffic was involved.	Used Metasploitable 2 as the remote authorized target and verified port 22.
HTTP connectivity worked, but the custom Snort HTTP alert did not reliably trigger.	Recorded HTTP as tested but did not claim it as a verified Snort alert.
ausearch -m USER_LOGIN -ts recent returned no matches.	Documented the result honestly and used verified USER_AUTH evidence.

8. Training Completion Evidence

The supplied screenshots document completed LetsDefend lessons. The report includes the strongest set of completion evidence rather than repeating every similar screen.

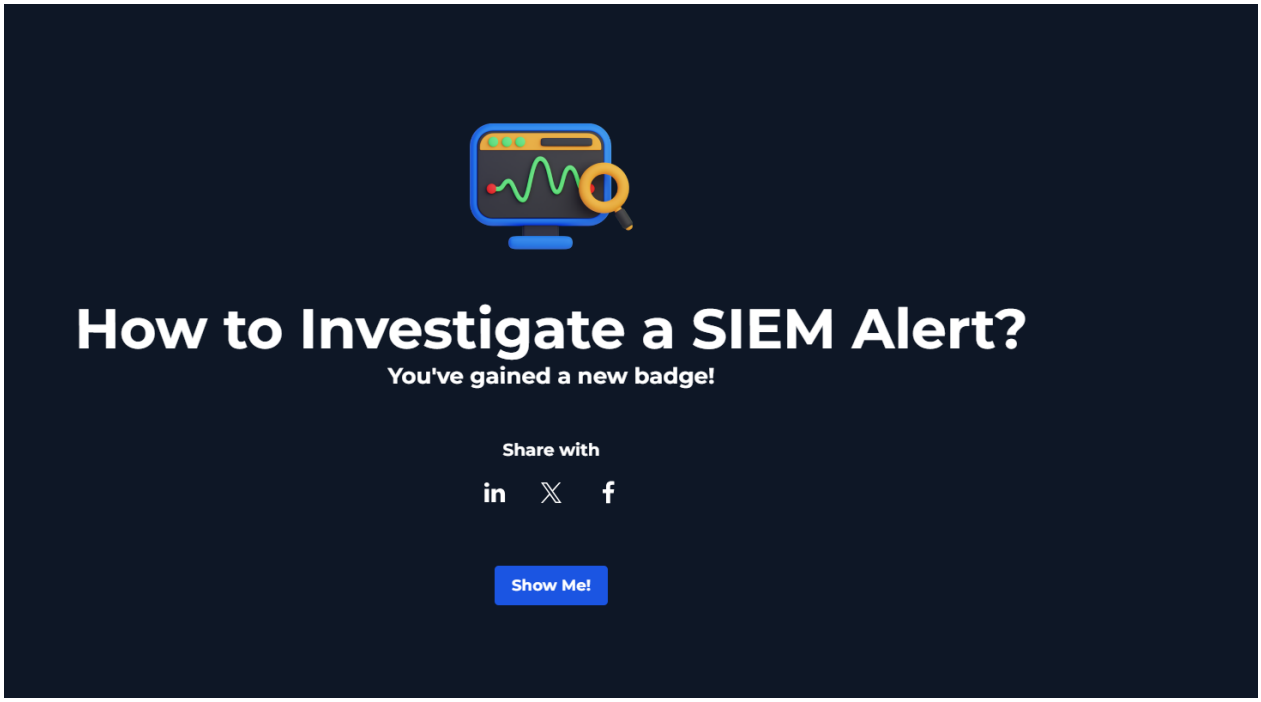


Figure 12. LetsDefend — How to Investigate a SIEM Alert.

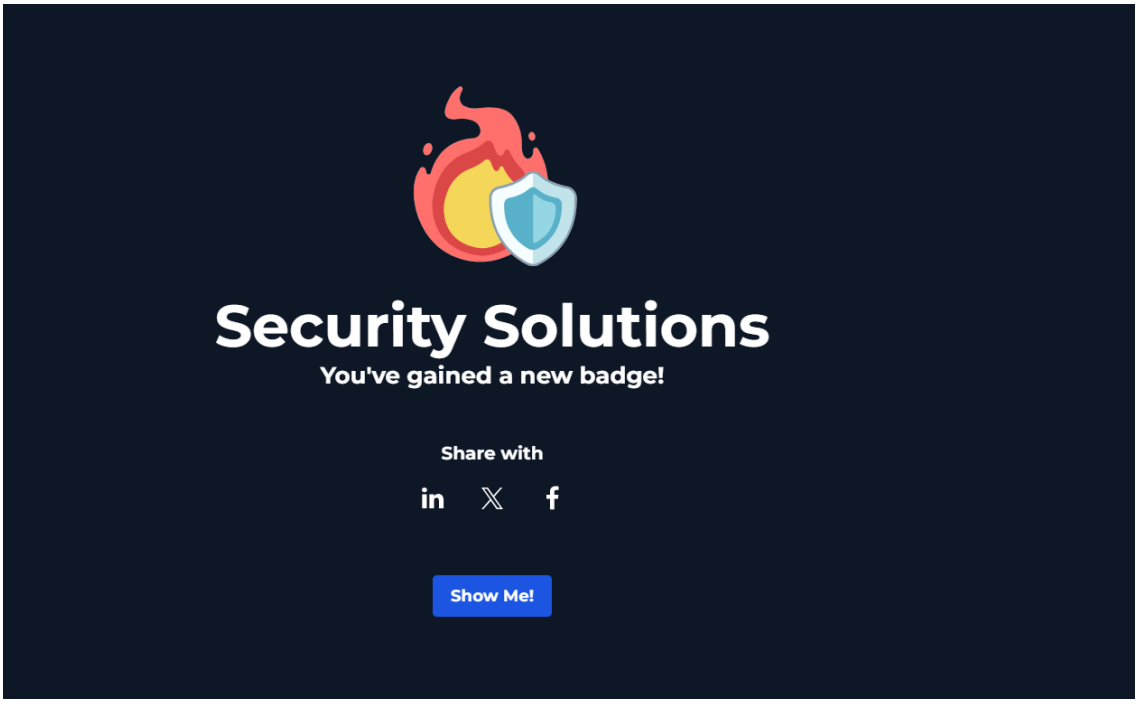


Figure 13. LetsDefend — Security Solutions.

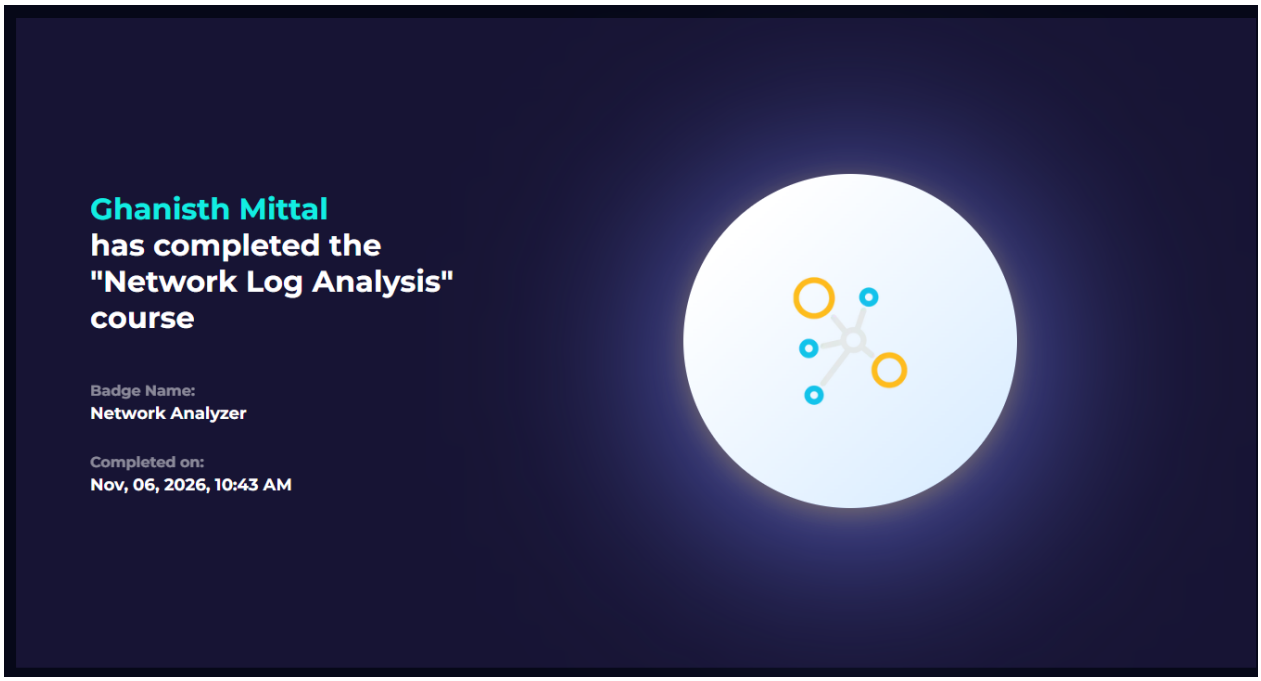


Figure 14. LetsDefend — Network Log Analysis.

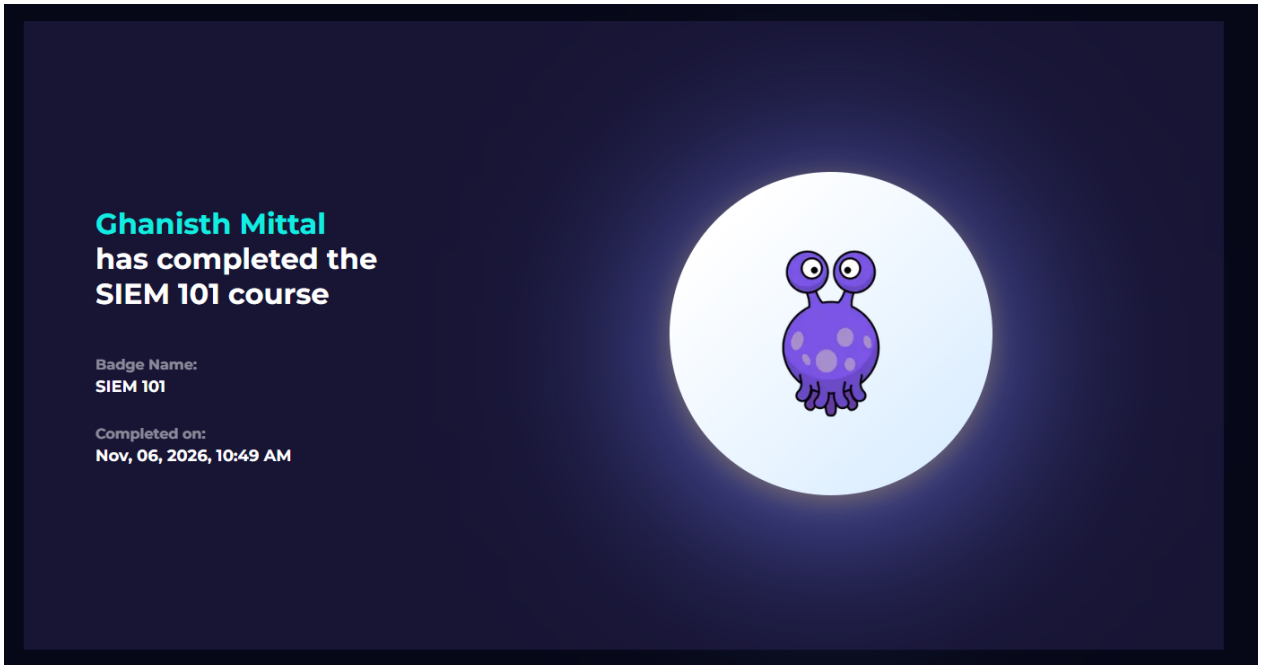


Figure 15. LetsDefend — SIEM 101.

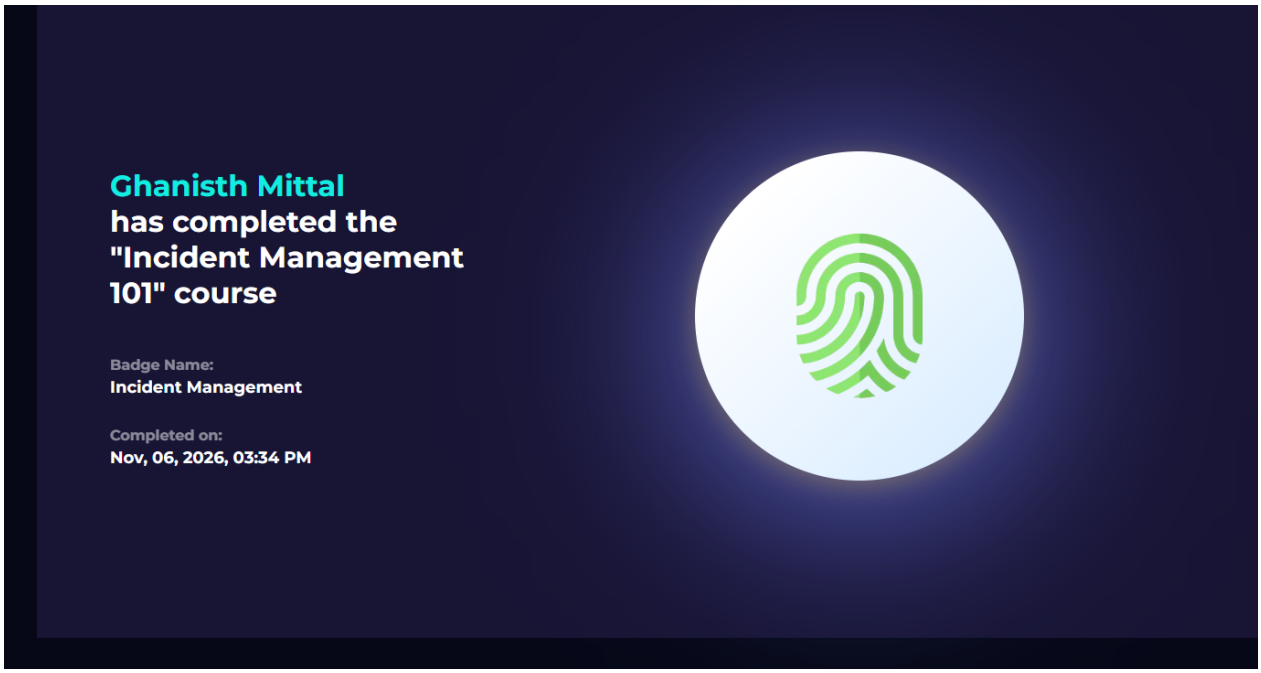


Figure 16. LetsDefend — Incident Management.

The supplied screenshot set also includes the completed training evidence for the remaining Week 3 lessons. The final submission should retain all completion screenshots in the LetsDefend findings folder.

9. Training Summary

TryHackMe — 4/4: Linux Logging for SOC; Intro to Logs; Intro to Log Analysis; Snort.

LetsDefend — 6/6: How to Investigate a SIEM Alert; Security Solutions; Network Log Analysis; SIEM 101; Incident Management; Incident Response on Linux.

10. Evidence Index

Folder	Evidence
Findings/Sysmon	Sysmon_Event_1.png; Sysmon_Event_3.png; Sysmon_Event_11.png; Sysmon_Operational_Log.png; Sysmon_User_Creation.png
Findings/auditd	Auditd_Command_Execution.png; auditd_EXECVE_Evidence.png; Auditd_ID_Command_Event.png; Auditd_Service_Status.png; auditd_USER_AUTH_Evidence.png
Findings/Snort	Snort_Configuration_Validation.png; Snort_Custom_Rules_Validation.png; Snort_ICMP_Alert.png; Snort_Installation_and_Network.png; Snort_Nmap.png; Snort_Nmap_T1046_Alert.png; Snort_SSH_Alert_ICMP_Metasploitable_Alert.png
Findings/TryHackMe	Completion screenshots for all 4 rooms
Findings/LetsDefend	Completion screenshots for all 6 lessons
Root	Week_3_SOC_Cheat_Sheet.docx

11. LinkedIn Submission

The Week 3 LinkedIn post was published. The internship submission should contain the **actual post URL**, not only the LinkedIn profile URL.

LinkedIn post URL: _____

12. Week 3 Reflection

The main learning was that SOC work is evidence-driven. Configuration validation alone is not enough: a detection should be tested with controlled activity and the resulting telemetry should be captured. The week also demonstrated why tool versions and traffic direction matter. The Snort threshold error, ICMP direction adjustment, remote SSH testing and missing USER_LOGIN event were handled through verification rather than assumptions.

13. Conclusion

Week 3 strengthened practical skills in endpoint logging, Linux auditing, IDS rule creation, reconnaissance detection and SIEM-style investigation. The report records specific commands, actual errors and fixes, and distinguishes verified evidence from tests that did not produce the expected alert.